Lesson 55: REINFORCE and Baselines

AI/ML Mathematics

Contents

1 Motivation and Overview 2

2 References 3

3 The REINFORCE Estimator 3

4 Baseline Subtraction and Variance Reduction 4

5 Variance Analysis and the Optimal Baseline 5

6 The REINFORCE Algorithms 7

7 Convergence of REINFORCE 8

8 Worked Example: Two-State MDP 9

9 Exercises 9

10 Summary 10

11 Further Reading 10

1 Motivation and Overview

The policy gradient theorem (Lesson 54) provides a clean expression for $\nabla_\theta J(\theta)$, but it involves the unknown action-value function $Q^\pi(s, a)$ and an expectation over the on-policy state distribution. To turn this into a practical algorithm, we need to estimate the gradient from sampled trajectories. The simplest approach is to replace $Q^\pi(s_t, a_t)$ with the sample return G_t – the actual discounted sum of rewards observed from time t onward. This yields the REINFORCE algorithm of Williams [1992], one of the earliest and most influential policy gradient methods.

However, REINFORCE in its basic form suffers from high variance: the sampled returns G_t fluctuate wildly across episodes, leading to noisy gradient estimates and slow convergence. This lesson develops the theory of *baseline subtraction*, showing that we can subtract any state-dependent function $b(s)$ from the return without biasing the gradient, and derives the *optimal baseline* that minimizes the variance of the gradient estimator. The tools developed here will let us understand the variance-bias tradeoff that governs all modern policy gradient algorithms.

RL/ML Connection

- **REINFORCE** [Williams, 1992]: The canonical Monte Carlo policy gradient algorithm derived directly from the policy gradient theorem with sampled returns.
- **Advantage actor-critic (A2C)**: Uses a learned value function $\hat{V}(s)$ as a baseline, reducing REINFORCE to updates proportional to the estimated advantage.
- **Generalized Advantage Estimation (GAE)** [Schulman et al., 2016]: Extends the baseline idea by interpolating between Monte Carlo returns and bootstrapped estimates via an exponentially weighted average.
- **PPO** [Schulman et al., 2017]: Builds on the REINFORCE gradient structure with a clipped surrogate objective for stable large-scale training.

Prerequisites. The reader must know the policy gradient theorem in both the log-derivative and expectation form (Lesson 54, Theorem 5.1), the baseline invariance proposition (Lesson 54, Proposition 7.1), score function properties (Lesson 54, Lemmas 4.1–4.2), and the definition of the discounted state visitation distribution (Lesson 54, Definition 3.3). From probability, we need the law of iterated expectation (tower property, Lesson 11), variance decomposition, and conditional independence in Markov chains (Lesson 9). Familiarity with stochastic approximation convergence conditions (Lesson 19, Robbins-Monro theorem) is helpful for the convergence section.

Learning objectives.

- **Derive** the REINFORCE gradient estimator from the policy gradient theorem and prove its unbiasedness.
- **Prove** that subtracting a state-dependent baseline preserves unbiasedness of the gradient estimate.
- **Derive** the optimal baseline that minimizes the variance of the REINFORCE gradient estimator.
- **Explain** why $V^\pi(s)$ is a near-optimal baseline and how it gives rise to the advantage function formulation.
- **State** convergence conditions for REINFORCE and justify the requirement of diminishing step sizes.

- **Compute** REINFORCE updates for concrete policy parameterizations on simple MDPs.

2 References

- Williams [1992] – The original REINFORCE paper; derives the algorithm and the baseline result.
- Sutton and Barto [2018, Chapter 13] – REINFORCE with and without baseline, intuitive presentation with pseudocode.
- Sutton et al. [2000] – Policy gradient theorem; the theoretical basis for REINFORCE.
- Schulman et al. [2016] – Generalized advantage estimation; extends the baseline idea to multi-step returns.
- Bertsekas and Tsitsiklis [1996, Chapter 6] – Convergence of stochastic gradient methods in approximate dynamic programming.
- Konda and Tsitsiklis [2003] – Actor-critic convergence; formalizes the connection between REINFORCE with baseline and actor-critic methods.

3 The REINFORCE Estimator

We begin by showing how to turn the policy gradient theorem into a practical Monte Carlo estimator using sampled trajectories.

Definition 3.1 (Trajectory and Return)**.** A *trajectory* $\tau = (S_0, A_0, R_1, S_1, A_1, R_2, \ldots)$ is generated by following π_θ: $A_t \sim \pi_\theta(\cdot \mid S_t)$, $S_{t+1} \sim \mathcal{P}(\cdot \mid S_t, A_t)$, $R_{t+1} = r(S_t, A_t)$. The *return from time* t is

$$G_t = \sum_{k=0}^{\infty} \gamma^k R_{t+k+1} = \sum_{k=0}^{\infty} \gamma^k r(S_{t+k}, A_{t+k}).$$

For episodic tasks terminating at time T, $G_t = \sum_{k=0}^{T-t-1} \gamma^k R_{t+k+1}$.

Lemma 3.2 (Return as Unbiased Estimate of Q^π)**.** *For any $t \geq 0$, conditioning on the state-action pair (S_t, A_t):*

$$\mathbb{E}[G_t \mid S_t = s, A_t = a] = Q^\pi(s, a).$$

Proof. By definition, $Q^\pi(s,a) = \mathbb{E}_{\pi_\theta}[\sum_{k=0}^{\infty} \gamma^k r(S_{t+k}, A_{t+k}) \mid S_t = s, A_t = a]$, and by the Markov property, this expectation depends only on (s, a) and is exactly $\mathbb{E}[G_t \mid S_t = s, A_t = a]$. □

Theorem 3.3 (Unbiasedness of REINFORCE)**.** *Let τ be a trajectory generated by π_θ starting from $S_0 = s_0$. The REINFORCE gradient estimator*

$$\hat{g}(\tau) = \sum_{t=0}^{\infty} \gamma^t \nabla_\theta \log \pi_\theta(A_t \mid S_t)\, G_t$$

satisfies $\mathbb{E}[\hat{g}(\tau)] = \nabla_\theta J(\theta)$, i.e., it is an unbiased estimator of the policy gradient.

Proof. We show that the expectation of $\hat{g}(\tau)$ equals the policy gradient derived in Lesson 54 (Theorem 5.1).

Step 1. Consider a single term at time t. By the tower property (conditioning on the history up to time t):

$$\begin{aligned}\mathbb{E}\big[\nabla_\theta \log \pi_\theta(A_t \mid S_t)\, G_t\big] &= \mathbb{E}\Big[\mathbb{E}\big[\nabla_\theta \log \pi_\theta(A_t \mid S_t)\, G_t \mid S_t, A_t\big]\Big] \\ &= \mathbb{E}\big[\nabla_\theta \log \pi_\theta(A_t \mid S_t)\, Q^\pi(S_t, A_t)\big],\end{aligned}$$

where we used Lemma 3.2 and the fact that $\nabla_\theta \log \pi_\theta(A_t \mid S_t)$ is measurable with respect to (S_t, A_t).

Step 2. The inner expectation over $A_t \sim \pi_\theta(\cdot \mid S_t)$ gives:

$$\mathbb{E}_{A_t}\big[\nabla_\theta \log \pi_\theta(A_t \mid S_t)\, Q^\pi(S_t, A_t) \mid S_t = s\big] = \sum_a \pi_\theta(a \mid s)\, \nabla_\theta \log \pi_\theta(a \mid s)\, Q^\pi(s,a) = \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi(s,a).$$

Step 3. Taking the outer expectation over S_t:

$$\mathbb{E}\big[\nabla_\theta \log \pi_\theta(A_t \mid S_t)\, G_t\big] = \sum_s \mathbb{P}(S_t = s \mid s_0, \pi_\theta) \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi(s,a).$$

Step 4. Summing over t with weights γ^t:

$$\begin{aligned}\mathbb{E}[\hat{g}(\tau)] &= \sum_{t=0}^{\infty} \gamma^t \sum_s \mathbb{P}(S_t = s \mid s_0, \pi_\theta) \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi(s,a) \\ &= \sum_s \underbrace{\left(\sum_{t=0}^{\infty} \gamma^t\, \mathbb{P}(S_t = s \mid s_0, \pi_\theta)\right)}_{\tilde{d}^{\pi_\theta}(s)} \sum_a \nabla_\theta \pi_\theta(a \mid s)\, Q^\pi(s,a) = \nabla_\theta J(\theta). \qquad \square\end{aligned}$$

REINFORCE Gradient Estimator

Given a trajectory $\tau = (S_0, A_0, R_1, S_1, A_1, R_2, \ldots)$ sampled under π_θ, the gradient of $J(\theta)$ can be estimated as

$$\hat{g}(\tau) = \sum_{t=0}^{T-1} \gamma^t\, \nabla_\theta \log \pi_\theta(A_t \mid S_t)\, G_t,$$

where $G_t = \sum_{k=0}^{T-t-1} \gamma^k R_{t+k+1}$ is the discounted return from time t and T is the episode length. This estimator is unbiased: $\mathbb{E}[\hat{g}(\tau)] = \nabla_\theta J(\theta)$.

Remark 3.4 (Causality). Past rewards $r(S_{t'}, A_{t'})$ for $t' < t$ can be dropped from the product with $\nabla_\theta \log \pi_\theta(A_t \mid S_t)$ because $\mathbb{E}[\nabla_\theta \log \pi_\theta(A_t \mid S_t) \mid S_t] = 0$ (Lesson 54, Lemma 4.2). This justifies using G_t rather than G_0.

Example 3.5 (REINFORCE on a Single-State Bandit)**.** Consider a bandit (single state s) with two actions $\mathcal{A} = \{0, 1\}$ and rewards $r(s, 0) = 1$, $r(s, 1) = 3$. Use a softmax policy: $\pi_\theta(1 \mid s) = \sigma(\theta) = 1/(1 + e^{-\theta})$, $\pi_\theta(0 \mid s) = 1 - \sigma(\theta)$. Then:

$$\nabla_\theta \log \pi_\theta(a \mid s) = \begin{cases} 1 - \sigma(\theta), & a = 1, \\ -\sigma(\theta), & a = 0. \end{cases}$$

The REINFORCE update $\theta \leftarrow \theta + \alpha\, r \cdot \nabla_\theta \log \pi_\theta(a \mid s)$ gives $\Delta\theta = 3\alpha(1 - \sigma(\theta)) > 0$ when $a = 1$, and $\Delta\theta = -\alpha\sigma(\theta) < 0$ when $a = 0$; both push θ toward selecting action 1. The true gradient $\nabla_\theta J = 2\sigma(\theta)(1 - \sigma(\theta)) > 0$ confirms this.

4 Baseline Subtraction and Variance Reduction

While REINFORCE is unbiased, its variance can be prohibitively high. We now develop the theory of baseline subtraction as a variance reduction technique.

Definition 4.1 (Baseline)**.** A *baseline* is any function $b : \mathcal{S} \to \mathbb{R}$ that depends on the state but not on the action. The *REINFORCE with baseline* estimator is

$$\hat{g}_b(\tau) = \sum_{t=0}^{T-1} \gamma^t\, \nabla_\theta \log \pi_\theta(A_t \mid S_t)\, \big(G_t - b(S_t)\big).$$

Proposition 4.2 (Unbiasedness with Baseline)**.** *For any baseline $b : \mathcal{S} \to \mathbb{R}$, the estimator $\hat{g}_b(\tau)$ is unbiased:* $\mathbb{E}[\hat{g}_b(\tau)] = \nabla_\theta J(\theta)$.

Proof. It suffices to show that $\mathbb{E}[\gamma^t \nabla_\theta \log \pi_\theta(A_t \mid S_t)\, b(S_t)] = 0$ for each t. By the tower property:

$$\begin{aligned}\mathbb{E}\big[\nabla_\theta \log \pi_\theta(A_t \mid S_t)\, b(S_t)\big] &= \mathbb{E}\Big[\mathbb{E}\big[\nabla_\theta \log \pi_\theta(A_t \mid S_t)\, b(S_t) \mid S_t\big]\Big] \\ &= \mathbb{E}\Big[b(S_t)\, \mathbb{E}\big[\nabla_\theta \log \pi_\theta(A_t \mid S_t) \mid S_t\big]\Big] \\ &= \mathbb{E}\big[b(S_t) \cdot 0\big] = 0,\end{aligned}$$

where we used that $b(S_t)$ is measurable with respect to S_t and $\mathbb{E}_{A_t \sim \pi_\theta}[\nabla_\theta \log \pi_\theta(A_t \mid S_t) \mid S_t] = 0$ by Lemma 4.2 of Lesson 54. Since the baseline term has zero expectation, $\mathbb{E}[\hat{g}_b] = \mathbb{E}[\hat{g}] = \nabla_\theta J(\theta)$. □

Definition 4.3 (Advantage Function)**.** The *advantage function* under policy π is

$$A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s).$$

It measures how much better action a is compared to the average action under π in state s. By definition, $\mathbb{E}_{a \sim \pi(\cdot \mid s)}[A^\pi(s, a)] = 0$.

Proposition 4.4 (Policy Gradient via Advantages)**.** *Using $b(s) = V^\pi(s)$ as the baseline, the policy gradient becomes*

$$\nabla_\theta J(\theta) = \mathbb{E}_{s \sim d^\pi,\, a \sim \pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, A^\pi(s, a)\big].$$

Proof. By the policy gradient theorem (Lesson 54, Theorem 5.1) and baseline invariance (Lesson 54, Proposition 7.1):

$$\nabla_\theta J(\theta) = \mathbb{E}_{s \sim d^\pi,\, a \sim \pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, (Q^\pi(s, a) - V^\pi(s))\big] = \mathbb{E}_{s \sim d^\pi,\, a \sim \pi_\theta}\big[\nabla_\theta \log \pi_\theta(a \mid s)\, A^\pi(s, a)\big].$$

□

RL/ML Connection

The advantage form is used in virtually all modern policy gradient algorithms. The advantage $A^\pi(s, a)$ is positive for actions better than average and negative for worse-than-average actions. The gradient update therefore increases the probability of good actions and decreases the probability of bad actions, providing a clearer learning signal than using the raw return G_t, which is always positive and pushes all actions in the gradient direction (with varying magnitude).

5 Variance Analysis and the Optimal Baseline

We now analyze the variance of the REINFORCE estimator and derive the baseline that minimizes it.

Definition 5.1 (Per-Step Gradient Estimator)**.** To simplify the variance analysis, consider the *per-step* gradient estimator at a single state s:

$$\hat{g}_b(s) = \nabla_\theta \log \pi_\theta(A \mid s)\, \big(Q^\pi(s, A) - b(s)\big), \quad A \sim \pi_\theta(\cdot \mid s).$$

We analyze the variance of each component $[\hat{g}_b(s)]_i$ for $i = 1, \ldots, d$, where d is the dimension of θ.

Remark 5.2. We use $Q^\pi(s, A)$ rather than G_t in the variance analysis to isolate the effect of the baseline. In practice, G_t introduces additional variance from the stochasticity of future transitions, but the baseline $b(s)$ cannot reduce that component.

Lemma 5.3 (Variance Decomposition)**.** *For the per-step estimator, the variance of the i-th component is*

$$\mathrm{Var}\big([\hat{g}_b(s)]_i\big) = \mathbb{E}_{A\sim\pi_\theta}\Big[\big(\partial_i \log \pi_\theta(A \mid s)\big)^2 \big(Q^\pi(s, A) - b(s)\big)^2\Big] - \big([\nabla_\theta J_s]_i\big)^2,$$

where $\partial_i = \partial/\partial\theta_i$ and $J_s = \sum_a \pi_\theta(a \mid s)\, Q^\pi(s, a)$. Since the second term is independent of b, minimizing the variance reduces to minimizing the first term.

Proof. This follows directly from $\mathrm{Var}(X) = \mathbb{E}[X^2] - (\mathbb{E}[X])^2$ and the fact that $\mathbb{E}[\hat{g}_b(s)] = \sum_a \nabla_\theta \pi_\theta(a \mid s) Q^\pi(s, a)$ is independent of b (by baseline invariance). □

Theorem 5.4 (Optimal Baseline – Scalar Case)**.** *For the i-th component of the gradient, the baseline that minimizes $\mathrm{Var}([\hat{g}_b(s)]_i)$ is*

$$b_i^*(s) = \frac{\mathbb{E}_{A\sim\pi_\theta}\Big[\big(\partial_i \log \pi_\theta(A \mid s)\big)^2 Q^\pi(s, A)\Big]}{\mathbb{E}_{A\sim\pi_\theta}\Big[\big(\partial_i \log \pi_\theta(A \mid s)\big)^2\Big]}.$$

This is a weighted average of $Q^\pi(s, A)$, where the weights are the squared score function components.

Proof. We minimize over $b(s)$ the quantity (suppressing the state argument):

$$f(b) = \mathbb{E}_A\big[(\partial_i \log \pi)^2 (Q(A) - b)^2\big] = \mathbb{E}_A\big[(\partial_i \log \pi)^2 Q(A)^2\big] - 2b\, \mathbb{E}_A\big[(\partial_i \log \pi)^2 Q(A)\big] + b^2\, \mathbb{E}_A\big[(\partial_i \log \pi)^2\big].$$

This is quadratic in b with positive leading coefficient (assuming the score is not identically zero). Setting $f'(b) = 0$:

$$-2\, \mathbb{E}_A[(\partial_i \log \pi)^2 Q(A)] + 2b\, \mathbb{E}_A[(\partial_i \log \pi)^2] = 0,$$

which gives $b_i^*(s)$ as stated. □

Since the optimal baseline differs for each component i, a single scalar baseline cannot simultaneously minimize all components. However, a practical and commonly used choice is $b(s) = V^\pi(s)$.

Theorem 5.5 (Optimal Baseline – Total Variance)**.** *The baseline minimizing the total variance $\sum_{i=1}^d \mathrm{Var}([\hat{g}_b(s)]_i)$ is*

$$b^*(s) = \frac{\mathbb{E}_{A\sim\pi_\theta}\big[\|\nabla_\theta \log \pi_\theta(A \mid s)\|^2 Q^\pi(s, A)\big]}{\mathbb{E}_{A\sim\pi_\theta}[\|\nabla_\theta \log \pi_\theta(A \mid s)\|^2]}.$$

Proof. The total variance is $\sum_{i=1}^d \mathrm{Var}([\hat{g}_b(s)]_i)$. By Lemma 5.3, we minimize:

$$\sum_{i=1}^d \mathbb{E}_A\big[(\partial_i \log \pi)^2 (Q(A) - b)^2\big] = \mathbb{E}_A\big[\|\nabla_\theta \log \pi\|^2 (Q(A) - b)^2\big].$$

This is quadratic in b with the same structure as the scalar case. Setting the derivative to zero yields $b^*(s)$ as stated. □

Optimal Baseline

The variance-minimizing baseline for the REINFORCE estimator is

$$b^*(s) = \frac{\mathbb{E}\big[\|\nabla_\theta \log \pi_\theta(A \mid s)\|^2 \, Q^\pi(s, A)\big]}{\mathbb{E}\big[\|\nabla_\theta \log \pi_\theta(A \mid s)\|^2\big]}.$$

When $\|\nabla_\theta \log \pi_\theta(a \mid s)\|^2$ is approximately constant across actions (which holds, e.g., for Gaussian policies with fixed variance in the small-variance limit), $b^*(s) \approx \mathbb{E}_A[Q^\pi(s, A)] = V^\pi(s)$. This justifies using $V^\pi(s)$ as a practical near-optimal baseline.

Example 5.6 (Optimal Baseline for a Bandit)**.** Revisiting Example 3.3 from Lesson 54 with the softmax policy on $\mathcal{A} = \{0, 1\}$: let $p = \sigma(\theta)$, so $\pi_\theta(1) = p$ and $\pi_\theta(0) = 1 - p$. The score functions are $\nabla_\theta \log \pi_\theta(1) = 1 - p$ and $\nabla_\theta \log \pi_\theta(0) = -p$. Then:

$$b^* = \frac{p(1-p)^2 \cdot r(1) + (1-p)p^2 \cdot r(0)}{p(1-p)^2 + (1-p)p^2} = \frac{p(1-p)[(1-p)r(1) + p \cdot r(0)]}{p(1-p)} = (1-p)r(1) + p \cdot r(0).$$

This equals $\mathbb{E}[r(A)] = V^\pi(s)$ exactly. For a bandit with two actions and a one-dimensional parameter, the optimal baseline and the value function coincide.

Optimal Baseline $\neq$ Value Function in General

The coincidence in Example 5.6 does not hold in general. When the squared score norm $\|\nabla_\theta \log \pi_\theta(a \mid s)\|^2$ varies significantly across actions, the optimal baseline $b^*(s)$ can differ substantially from $V^\pi(s)$. Nevertheless, $V^\pi(s)$ remains the standard practical choice because (a) it is much simpler to estimate, (b) the suboptimality in variance is typically small, and (c) it leads naturally to the advantage function.

6 The REINFORCE Algorithms

We now state the complete REINFORCE algorithms in both the basic and baseline variants.

Definition 6.1 (REINFORCE Update Rule)**.** The *REINFORCE* parameter update at iteration k using a single episode $\tau_k = (S_0, A_0, R_1, \ldots, S_{T-1}, A_{T-1}, R_T)$ is

$$\theta_{k+1} = \theta_k + \alpha_k \sum_{t=0}^{T-1} \gamma^t \, \nabla_\theta \log \pi_{\theta_k}(A_t \mid S_t) \, G_t,$$

where $\alpha_k > 0$ is the step size (learning rate).

Definition 6.2 (REINFORCE with Baseline Update Rule)**.** Given a baseline function $b : \mathcal{S} \to \mathbb{R}$ (typically an estimate $\hat{V}(s)$ of $V^\pi(s)$), the update is

$$\theta_{k+1} = \theta_k + \alpha_k \sum_{t=0}^{T-1} \gamma^t \, \nabla_\theta \log \pi_{\theta_k}(A_t \mid S_t) \big(G_t - b(S_t)\big).$$

When $b(s) = \hat{V}_w(s)$ is parameterized by w, the baseline parameters are updated simultaneously:

$$w_{k+1} = w_k + \beta_k \sum_{t=0}^{T-1} \gamma^t \, (G_t - \hat{V}_{w_k}(S_t)) \, \nabla_w \hat{V}_{w_k}(S_t).$$

RL/ML Connection

The two-parameter update (for θ and w) is the simplest example of an *actor-critic* architecture. The "actor" is the policy π_θ and the "critic" is the value estimate $\hat{V}_w$. REINFORCE with a learned baseline can be viewed as a Monte Carlo actor-critic: the critic uses full returns G_t rather than bootstrapped TD estimates. Replacing G_t with a TD target $R_{t+1} + \gamma \hat{V}_w(S_{t+1})$ yields one-step actor-critic methods, which trade increased bias for reduced variance.

7 Convergence of REINFORCE

We now establish conditions under which REINFORCE converges to a stationary point of $J(\theta)$.

Assumption 7.1 (Regularity Conditions)**.** We assume:

(i) The parameter space $\Theta \subseteq \mathbb{R}^d$ is open.

(ii) $\pi_\theta(a \mid s) > 0$ for all s, a, θ and is twice continuously differentiable in θ.

(iii) The score function satisfies $\mathbb{E}_{s\sim d^\pi, a\sim\pi_\theta}[\|\nabla_\theta \log \pi_\theta(a \mid s)\|^2 \cdot Q^\pi(s,a)^2] < \infty$ for all θ.

(iv) $J(\theta)$ is bounded above: $\sup_\theta J(\theta) < \infty$.

(v) $\nabla_\theta J(\theta)$ is L-Lipschitz continuous.

Assumption 7.2 (Step Size Schedule)**.** The step sizes $\{\alpha_k\}_{k\geq 0}$ satisfy the Robbins-Monro conditions:

$$\sum_{k=0}^{\infty} \alpha_k = \infty, \qquad \sum_{k=0}^{\infty} \alpha_k^2 < \infty.$$

Theorem 7.3 (Convergence of REINFORCE)**.** *Under Assumptions 7.1 and 7.2, the REINFORCE iterates $\{\theta_k\}$ satisfy*

$$\lim_{k\to\infty} \nabla_\theta J(\theta_k) = 0 \quad a.s.$$

That is, θ_k converges almost surely to the set of stationary points of $J(\theta)$.

Proof. Write $\theta_{k+1} = \theta_k + \alpha_k(\nabla_\theta J(\theta_k) + \xi_k)$ where $\xi_k = \hat{g}(\tau_k) - \nabla_\theta J(\theta_k)$ satisfies $\mathbb{E}[\xi_k \mid \theta_k] = 0$ by Theorem 3.3.

Step 1. By Assumption 7.1(iii), $\mathbb{E}[\|\hat{g}(\tau_k)\|^2 \mid \theta_k] \leq C$ for some constant C.

Step 2. Define $V_k = \sup_\theta J(\theta) - J(\theta_k) \geq 0$. The L-Lipschitz gradient gives the descent inequality:

$$\mathbb{E}[V_{k+1} \mid \theta_k] \leq V_k - \alpha_k \|\nabla J(\theta_k)\|^2 + \frac{LC}{2}\alpha_k^2.$$

Step 3. Since $\sum \alpha_k^2 < \infty$, the Robbins-Siegmund theorem implies V_k converges a.s. and $\sum_k \alpha_k \|\nabla J(\theta_k)\|^2 < \infty$ a.s. Combined with $\sum \alpha_k = \infty$, this gives $\lim_{k\to\infty} \|\nabla J(\theta_k)\| = 0$ a.s. □

Local Optima

Theorem 7.3 guarantees convergence to a *stationary point*, not a global maximum. The policy objective $J(\theta)$ is generally non-concave, so REINFORCE may converge to a local maximum or saddle point. In practice, the choice of policy parameterization (e.g., neural networks) determines the loss landscape, and multiple restarts or entropy regularization are used to mitigate this issue.

8 Worked Example: Two-State MDP

We illustrate REINFORCE with baseline on a concrete MDP to build intuition.

Example 8.1 (Two-State MDP)**.** Consider an MDP with $\mathcal{S} = \{s_1, s_2\}$, $\mathcal{A} = \{\text{left}, \text{right}\}$, and $\gamma = 0.9$. The transitions and rewards are:

State	Action	Next State	Reward
s_1	left	s_1	$+1$
s_1	right	s_2	$+5$
s_2	left	s_1	0
s_2	right	s_2	$+2$

We use a softmax policy with $\pi_\theta(\text{right} \mid s_i) = \sigma(\theta_i)$ where $\theta = (\theta_1, \theta_2) \in \mathbb{R}^2$. The score for state s_i is $\nabla_{\theta_i} \log \pi_\theta(\text{right} \mid s_i) = 1 - \sigma(\theta_i)$ and $\nabla_{\theta_i} \log \pi_\theta(\text{left} \mid s_i) = -\sigma(\theta_i)$, with all cross-derivatives zero.

At $\theta = (0, 0)$ (equal action probabilities $p = 0.5$), solving the Bellman system $(I - \gamma P^\pi)V = r^\pi$ gives $V^\pi(s_1) \approx 17.39$ and $V^\pi(s_2) \approx 16.05$.

The action-values at s_1 are: $Q^\pi(s_1, \text{right}) = 5 + 0.9 \cdot 16.05 \approx 19.45$ and $Q^\pi(s_1, \text{left}) = 1 + 0.9 \cdot 17.39 \approx 16.65$. The advantages are $A^\pi(s_1, \text{right}) \approx 2.06$ and $A^\pi(s_1, \text{left}) \approx -0.74$.

The REINFORCE with baseline gradient at s_1 (component θ_1) is:

$$\nabla_{\theta_1} J \approx 0.5 \cdot 0.5 \cdot 2.06 + 0.5 \cdot (-0.5) \cdot (-0.74) = 0.515 + 0.185 = 0.70.$$

This positive gradient increases θ_1, making "right" more likely in s_1, which is correct since $Q^\pi(s_1, \text{right}) > Q^\pi(s_1, \text{left})$.

9 Exercises

Exercise 9.1 (Verify Unbiasedness)**.** $(*)$
For the bandit in Example 3.5 (Lesson 54 setup with $r(0) = 1$, $r(1) = 3$, $p = \sigma(\theta) = 0.5$), enumerate all possible one-step trajectories and their probabilities. Explicitly compute $\mathbb{E}[\hat{g}(\tau)]$ and verify it equals $\nabla_\theta J(\theta)$.

Exercise 9.2 (Variance Reduction Ratio)**.** $(*)$
Continuing with the bandit above, compute:

(a) $\mathrm{Var}(\hat{g})$ for REINFORCE without baseline,

(b) $\mathrm{Var}(\hat{g}_b)$ for REINFORCE with the optimal baseline $b^* = V^\pi$.

What is the ratio $\mathrm{Var}(\hat{g}_b)/\mathrm{Var}(\hat{g})$?

Exercise 9.3 (Baseline Does Not Reduce All Variance)**.** $(**)$
Show that even with the optimal baseline $b^*(s)$, the variance of the REINFORCE estimator is strictly positive as long as $Q^\pi(s, a)$ is not constant across actions. Specifically, prove that

$$\mathrm{Var}([\hat{g}_{b^*}(s)]_i) = 0 \quad \Longleftrightarrow \quad Q^\pi(s, a) = c \text{ for all } a \in \mathrm{supp}(\pi_\theta(\cdot \mid s)).$$

Hint: Use the Cauchy-Schwarz inequality to analyze when the variance equals zero.

Exercise 9.4 (REINFORCE with Discounted Weighting)**.** $(**)$
An alternative REINFORCE estimator uses the "undiscounted" form:

$$\hat{g}'(\tau) = \sum_{t=0}^{T-1} \nabla_\theta \log \pi_\theta(A_t \mid S_t)\, G_t,$$

without the γ^t factor. Show that this estimator is *not* an unbiased estimator of $\nabla_\theta J(\theta)$ but rather estimates $\nabla_\theta \tilde{J}(\theta)$ for a different objective $\tilde{J}$. Identify $\tilde{J}$ and explain why this alternative is still commonly used in practice.

Exercise 9.5 (Convergence Rate)**.** $(***)$
Under the conditions of Theorem 7.3, suppose $\alpha_k = c/(k + k_0)$ for constants $c > 0$ and $k_0 > 0$. Using the bound from Step 2 of the convergence proof, show that

$$\min_{0 \le k \le K} \mathbb{E}\big[\|\nabla_\theta J(\theta_k)\|^2\big] = O\left(\frac{\log K}{K}\right).$$

Hint: Sum the one-step descent inequality from $k = 0$ to K and bound the step-size sums.

Exercise 9.6 (Action-Dependent Baseline)**.** $(***)$
Show that if the baseline is allowed to depend on the action, i.e., $b(s, a)$, then the gradient estimator $\nabla_\theta \log \pi_\theta(a \mid s) \cdot (Q^\pi(s, a) - b(s, a))$ is generally *biased*. Derive the exact bias term and find a condition on $b(s, a)$ under which the bias vanishes.

10 Summary

- **REINFORCE** uses G_t as an unbiased estimate of $Q^\pi(S_t, A_t)$, yielding an unbiased Monte Carlo policy gradient estimator.
- **Baseline subtraction**: replacing G_t with $G_t - b(S_t)$ preserves unbiasedness for any state-dependent $b(s)$, by the zero-mean property of the score function.
- The **advantage function** $A^\pi(s, a) = Q^\pi(s, a) - V^\pi(s)$ arises from using $V^\pi(s)$ as baseline, centering the gradient signal around average action quality.
- The **optimal baseline** is a score-weighted average of Q^π and approximately equals $V^\pi(s)$ when the squared score norm is nearly action-independent.
- REINFORCE converges to a **stationary point** (not necessarily a global maximum) under Robbins-Monro step-size conditions.
- REINFORCE with a learned baseline $\hat{V}_w(s)$ is the simplest **actor-critic** architecture, bridging Monte Carlo and bootstrapped policy gradient methods.

11 Further Reading

- **Lesson 56** (Trust Region Methods) constrains the policy update to a KL-divergence neighborhood, addressing the step-size sensitivity of REINFORCE.
- **Lesson 57** (PPO) builds on the advantage formulation with a clipped surrogate objective for stable large-scale training.
- Schulman et al. [2016] introduces *Generalized Advantage Estimation* (GAE), interpolating between Monte Carlo and TD advantage estimates via a parameter λ.
- Konda and Tsitsiklis [2003] provides a rigorous two-timescale convergence analysis of actor-critic methods with a simultaneously learned critic.
- Bhatnagar et al. [2009] analyzes natural actor-critic algorithms, connecting baselines to Fisher information geometry.
- Bertsekas and Tsitsiklis [1996, Chapter 6] gives a deeper treatment of variance reduction in gradient estimation, including general control variates.

References

Dimitri P. Bertsekas and John N. Tsitsiklis. *Neuro-Dynamic Programming.* Athena Scientific, 1996.

Shalabh Bhatnagar, Richard S. Sutton, Mohammad Ghavamzadeh, and Mark Lee. Natural actor-critic algorithms. *Automatica*, 45(11):2471–2482, 2009.

Vijay R. Konda and John N. Tsitsiklis. On actor-critic algorithms. *SIAM Journal on Control and Optimization*, 42(4):1143–1166, 2003.

John Schulman, Philipp Moritz, Sergey Levine, Michael Jordan, and Pieter Abbeel. High-dimensional continuous control using generalized advantage estimation. *Proceedings of the International Conference on Learning Representations*, 2016.

John Schulman, Filip Wolski, Prafulla Dhariwal, Alec Radford, and Oleg Klimov. Proximal policy optimization algorithms. *arXiv preprint arXiv:1707.06347*, 2017.

Richard S. Sutton and Andrew G. Barto. *Reinforcement Learning: An Introduction.* MIT Press, 2nd edition, 2018.

Richard S. Sutton, David McAllester, Satinder Singh, and Yishay Mansour. Policy gradient methods for reinforcement learning with function approximation. *Advances in Neural Information Processing Systems*, 12:1057–1063, 2000.

Ronald J. Williams. Simple statistical gradient-following algorithms for connectionist reinforcement learning. *Machine Learning*, 8(3–4):229–256, 1992.